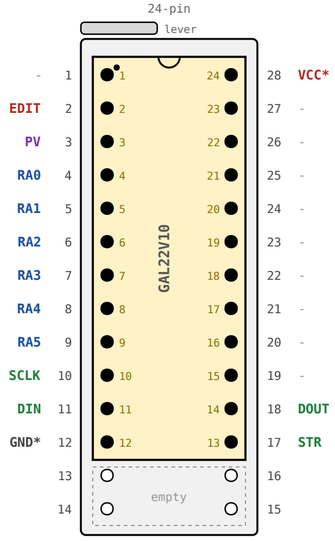
J48 ZIF-28 socket pin map — every device is head-aligned (chip pin 1 at ZIF pin 1)

20-pin parts use ZIF 1–10 and 19–28; 24-pin parts use ZIF 1–12 and 17–28; 28-pin parts fill the socket. GND and VCC are switched per size (Rev H).

22V10 / 20XV10 / ATF750C

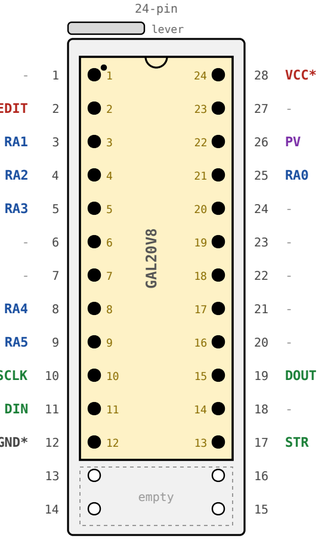
24-pin



* GND: ZIF 12 via Q4 (U9 Q4)
* VCC: ZIF 28 via Q6 (U9 Q6)

20V8

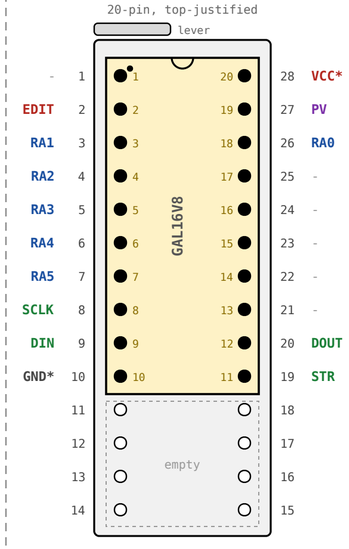
24-pin



* GND: ZIF 12 via Q4 (U9 Q4)
* VCC: ZIF 28 via Q6 (U9 Q6)

16V8 (incl. ATF16V8B)

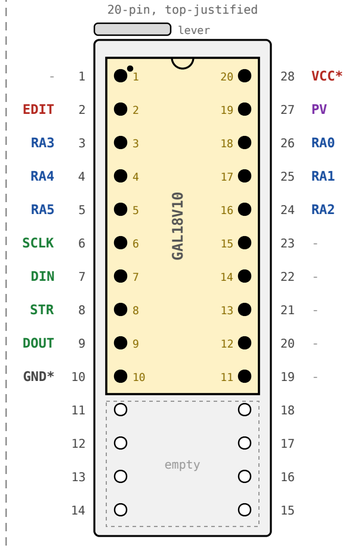
20-pin, top-justified



* GND: ZIF 10 via Q1 (U2 Q6)
* VCC: ZIF 28 via Q6 (U9 Q6)

18V10

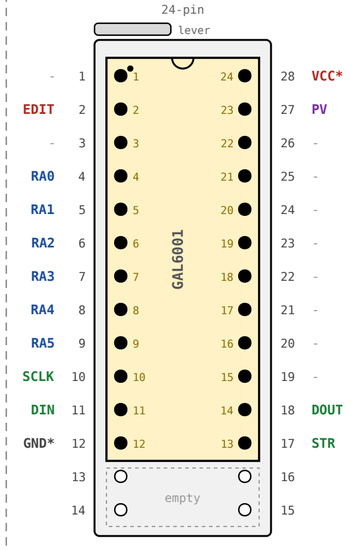
20-pin, top-justified



* GND: ZIF 10 via Q1 (U2 Q6)
* VCC: ZIF 28 via Q6 (U9 Q6)

6001 / 6002

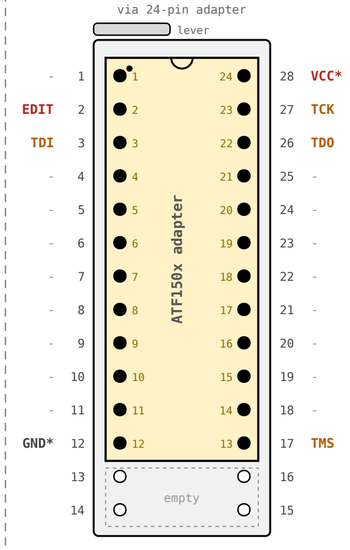
24-pin



* GND: ZIF 12 via Q4 (U9 Q4)
* VCC: ZIF 28 via Q6 (U9 Q6)

JTAG: ATF150x

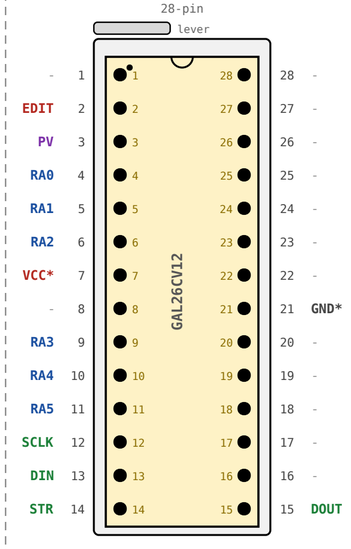
via 24-pin adapter



* GND: ZIF 12 via Q4 (U9 Q4)
* VCC: ZIF 28 via Q6 (U9 Q6)

26CV12 / 26V12

28-pin



* GND: ZIF 21 via Q5 (U9 Q5)
* VCC: ZIF 7 via Q7 (U9 Q7)

EDIT VPP / edit-mode supply (ZIF 2) **RA0** row address **SCLK** serial clock, DIN/DOUT data, STR strobe **PV** program / verify **TDI** JTAG **VCC** switched VCC_SKT (* = switch, see panel notes)

● chip pin present ○ socket contact unused

ZIF-28 pin map: each device's role at every ZIF pin, and the Rev H connection

ZIF	Rev H net	driven by	22V10 / 20XV10 / ATF750C 24-pin (role · chip pin)	20V8 24-pin (role · chip pin)	16V8 (incl. ATF16V8B) 20-pin, top-justified (role · chip pin)	18V10 20-pin, top-justified (role · chip pin)	6001 / 6002 24-pin (role · chip pin)	JTAG: ATF150x via 24-pin adapter (role · chip pin)	26CV12 / 26V12 28-pin
1	pull-down	R20 10k	— · 1	— · 1	— · 1	— · 1	— · 1	— · 1	—
2	VPP	VPP gen.	EDIT · 2	EDIT · 2	EDIT · 2	EDIT · 2	EDIT · 2	EDIT · 2	EDIT
3	U1 PD0	MCU	PV · 3	RA1 · 3	RA1 · 3	RA3 · 3	— · 3	TDI · 3	PV
4	U2 Q0	595	RA0 · 4	RA2 · 4	RA2 · 4	RA4 · 4	RA0 · 4	— · 4	RA0
5	U2 Q3	595	RA1 · 5	RA3 · 5	RA3 · 5	RA5 · 5	RA1 · 5	— · 5	RA1
6	U2 Q2	595	RA2 · 6	— · 6	RA4 · 6	SCLK · 6	RA2 · 6	— · 6	RA2
7	U2 Q1 + Q7	595 / VCC sw	RA3 · 7	— · 7	RA5 · 7	DIN · 7	RA3 · 7	— · 7	VCC (Q7)
8	U1 PD2	MCU	RA4 · 8	RA4 · 8	SCLK · 8	STR · 8	RA4 · 8	— · 8	—
9	U1 PD3	MCU	RA5 · 9	RA5 · 9	DIN · 9	DOUT · 9	RA5 · 9	— · 9	RA3
10	U1 PD4 + Q1	MCU / GND sw	SCLK · 10	SCLK · 10	GND · 10 (Q1)	GND · 10 (Q1)	SCLK · 10	— · 10	RA4
11	U1 PD5	MCU	DIN · 11	DIN · 11	(no chip pin)	(no chip pin)	DIN · 11	— · 11	RA5
12	U9 Q0 + Q4	595 / GND sw	GND · 12 (Q4)	GND · 12 (Q4)	(no chip pin)	(no chip pin)	GND · 12 (Q4)	GND · 12 (Q4)	SCLK
13	U9 Q1	595	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	DIN
14	U9 Q2	595	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	STR
15	U1 PA3	MCU	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	DOUT
16	pull-down	R46 10k	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	(no chip pin)	—
17	U1 PD6	MCU	STR · 13	STR · 13	(no chip pin)	(no chip pin)	STR · 13	TMS · 13	—
18	U1 PD7	MCU	DOUT · 14	— · 14	(no chip pin)	(no chip pin)	DOUT · 14	— · 14	—
19	U1 PC3	MCU	— · 15	DOUT · 15	STR · 11	— · 11	— · 15	— · 15	—
20	U1 PF0	MCU	— · 16	— · 16	DOUT · 12	— · 12	— · 16	— · 16	—
21	pull-down + Q5	R21 / GND sw	— · 17	— · 17	— · 13	— · 13	— · 17	— · 17	GND (Q5)
22	pull-down	R22 10k	— · 18	— · 18	— · 14	— · 14	— · 18	— · 18	—
23	pull-down	R23 10k	— · 19	— · 19	— · 15	— · 15	— · 19	— · 19	—
24	U2 Q5	595	— · 20	— · 20	— · 16	RA2 · 16	— · 20	— · 20	—
25	U2 Q4	595	— · 21	RA0 · 21	— · 17	RA1 · 17	— · 21	— · 21	—
26	U1 PF1	MCU	— · 22	PV · 22	RA0 · 18	RA0 · 18	— · 22	TDO · 22	—
27	U2 Q7	595	— · 23	— · 23	PV · 19	PV · 19	PV · 23	TCK · 23	—
28	VCC_SKT via Q6	VCC sw + R47	VCC · 24 (Q6)	VCC · 24 (Q6)	VCC · 20 (Q6)	VCC · 20 (Q6)	VCC · 24 (Q6)	VCC · 24 (Q6)	—

1. Role names: EDIT = VPP pin in edit mode, PV = program/verify, STR = strobe, DIN/DOUT = serial data in/out, RA0–RA5 = row address. '—' = held low or undriven.
2. Signals the MCU reads (DOUT, TDO) land on MCU pins: ZIF 9, 15, 18, 19, 20 and 26. ZIF 6 (18V10 SCLK), 12–14 (26CV12 SCLK, DIN, STR) and 27 (PV, JTAG TCK) go through a 74HC595 and change only on a latch pulse.
3. GND and VCC are switched per chip size and all switches are off at reset: one GND switch (Q1, Q4 or Q5) and one VCC switch (Q6 or Q7) at a time. Never Q6 with Q7; never Q4 with a 28-pin part.
4. ZIF 1, 16, 22, 23 and (when Q6 is off) 28 have pull-downs; ZIF 21's pull-down is shared with the Q5 ground switch.
5. Chip pin for 24-pin parts: ZIF 1–12 = pins 1–12, ZIF 17–28 = pins 13–24. For 20-pin parts: ZIF 1–10 = pins 1–10, ZIF 19–28 = pins 11–20.

J48 GAL / PAL Programmer — AVR16DU28
Rev H · ZIF-28 pin map · 2026-10-17
Pin roles: published device pinouts